

Lecture Three

1 The Continuity

1.1 Continuity at a point on an open interval

Informally, we might say that a function is continuous on an open interval if its graph can be drawn with a pencil without lifting the pencil from the paper. i.e. A function f is continuous at $x = c$ means there is no interruption in the graph of f at c . That is, its graph is unbroken at c and there are no holes, jumps, or gaps. The following figure identifies three values of x at which the graph of f is not continuous. At all other points in the interval (a, b) the graph of f is uninterrupted and **continuous**.

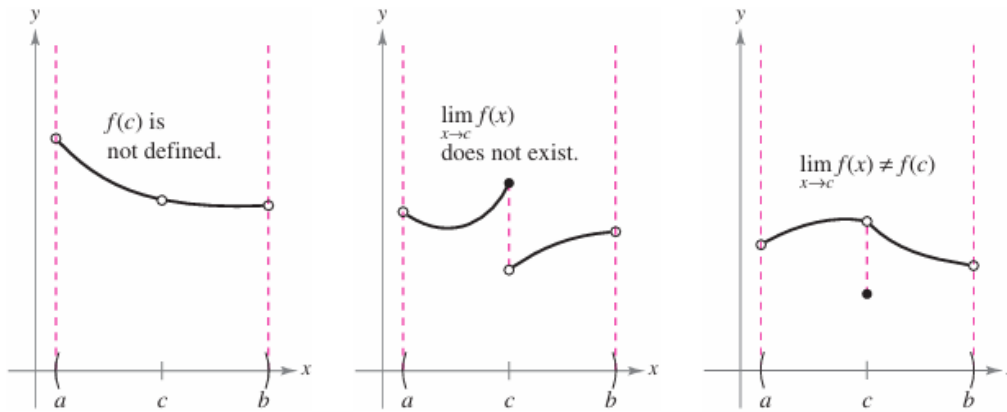


Figure 1: Three cases for the discontinuity

1.2 Concept of Continuity

Definition 1.2.1 Continuity at a point:

A function f is continuous at c if the following three conditions are met.

1. $f(c)$ is defined
2. $\lim_{x \rightarrow c} f(x)$ exists.
3. $f(c) = \lim_{x \rightarrow c} f(x)$

Continuity on an open interval:

A function is **continuous on an open interval** (a, b) if it is continuous at each point in the interval.

Definition 1.2.2 Let $f(x)$ be a **defined function** at $x = c$, we say that $f(x)$ is continuous at $x = c$ if the following condition satisfied.

$$\lim_{h \rightarrow 0} [f(c+h) - f(c)] = 0$$

Example 1.2.1 Given $f(x) = x^2$, prove that it is continuous on $\mathbb{R}$.

Proof

To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c \in \mathbb{R}$.

$$\begin{aligned} \lim_{h \rightarrow 0} [f(c+h) - f(c)] &= \lim_{h \rightarrow 0} [f(c+h)^2 - c^2] = \lim_{h \rightarrow 0} [c^2 + 2ch + h^2 - c^2] \\ &= \lim_{h \rightarrow 0} [h(2c+h)] = 0 \times (2c+0) = 0. \end{aligned}$$

$\Rightarrow f(x) = x^2$ is continuous at $c \Rightarrow$ continuous at $\mathbb{R}$.

Example 1.2.2 Prove that the function $f(x) = \cos ax$, continuous on $\mathbb{R}$.

Proof

To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c \in \mathbb{R}$.

$$\begin{aligned}\lim_{h \rightarrow 0} [f(c+h) - f(c)] &= \lim_{h \rightarrow 0} [\cos(ac+ah) - \cos(ac)] = -2 \lim_{h \rightarrow 0} \left[\sin\left(\frac{ac+ah+ac}{2}\right) \sin\left(\frac{ac+ah-ac}{2}\right) \right] \\ &= -2 \lim_{h \rightarrow 0} \left[\sin\left(ac + \frac{ah}{2}\right) \sin\left(\frac{ah}{2}\right) \right] = -2 \times \sin ac \times 0 = 0.\end{aligned}$$

$\Rightarrow f(x) = \cos ax$ is continuous at $c \Rightarrow$ continuous at $\mathbb{R}$.

Example 1.2.3 Prove that the function $f(x) = a^x$, $a > 0$ is continuous on $\mathbb{R}$.

Proof

To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c \in \mathbb{R}$.

$$\begin{aligned}\lim_{h \rightarrow 0} [f(c+h) - f(c)] &= \lim_{h \rightarrow 0} [a^{c+h} - a^c] = \lim_{h \rightarrow 0} [a^c (a^h - 1)] \\ &= a^c (1 - 1) = a^c \times 0 = 0\end{aligned}$$

$\Rightarrow f(x) = a^x$ is continuous on $\mathbb{R}$

Example 1.2.4 Prove that the function $f(x) = \log_a x$, $a > 0$ is continuous on $\mathbb{R}^+$.

Proof

To prove the function is continuous on $\mathbb{R}$, choose an arbitrary point $x = c$, s.t $c > 0, c \in \mathbb{R}$.

$$\begin{aligned}\lim_{h \rightarrow 0} [f(c+h) - f(c)] &= \lim_{h \rightarrow 0} [\log_a(c+h) - \log_a c] = \lim_{h \rightarrow 0} \left[\log_a \left(\frac{c+h}{c} \right) \right] \\ &= \lim_{h \rightarrow 0} \left[\log_a \left(1 + \frac{h}{c} \right) \right] = \log_a \left(\lim_{h \rightarrow 0} \left[1 + \frac{h}{c} \right] \right) = \log_a(1) = 0\end{aligned}$$

Example 1.2.5 Discuss the continuity of the function

$$f(x) = \begin{cases} 3x - 1 & , x < 1 \\ 2x^2 & , x \geq 1 \end{cases}$$

at $x = 1$.

Sol

Here we will study the three conditions on definition 1.2.1.

$$1. f(1) = 2(1)^2 = 2$$

2. For finding the limit:

$$\begin{aligned}\lim_{x \rightarrow 1^+} f(x) &= \lim_{\epsilon \rightarrow 0} f(a + \epsilon) = \lim_{\epsilon \rightarrow 0} 2(1 + \epsilon)^2 = 2(1 + 0)^2 = 2(1)^2 = 2. \\ \lim_{x \rightarrow 1^-} f(x) &= \lim_{\epsilon \rightarrow 0} f(a - \epsilon) = \lim_{\epsilon \rightarrow 0} 3(1 - \epsilon) - 1 = 3(1) - 1 = 3 - 1 = 2.\end{aligned}$$

$$\text{Since } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) \Rightarrow \lim_{x \rightarrow 1} f(x) \text{ exists and } = 2$$

3. Clearly $f(1) = \lim_{x \rightarrow 1} f(x)$, therefore the given function is continuous at $x = 1$.

Example 1.2.6 Determine whether the following function is continuous at $x = 0$.

$$f(x) = \begin{cases} \frac{\sin \pi x}{x} & , x < 0 \\ \frac{\sin \pi x}{x+1} & , x \geq 0 \end{cases}$$

Sol

$$1. f(1) = \frac{\sin(0\pi)}{0+1} = \frac{0}{1} = 0$$

2.

$$\begin{aligned}\lim_{x \rightarrow 0^+} f(x) &= \lim_{\epsilon \rightarrow 0} f(a + \epsilon) = \lim_{\epsilon \rightarrow 0} \frac{\sin(\pi(0 + \epsilon))}{0 + \epsilon + 1} = \lim_{\epsilon \rightarrow 0} \frac{\sin \pi \epsilon}{\epsilon + 1} = \frac{\sin 0}{0 + 1} = \frac{0}{1} = 0 \\ \lim_{x \rightarrow 0^-} f(x) &= \lim_{\epsilon \rightarrow 0} f(a - \epsilon) = \lim_{\epsilon \rightarrow 0} \frac{\sin(\pi(0 - \epsilon))}{0 - \epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\sin \pi \epsilon}{\epsilon} = \pi\end{aligned}$$

$\Rightarrow \lim_{x \rightarrow 0} f(x)$ not exist because $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$

Example 1.2.7 Find the constant such that the function is continuous on the entire real line.

1.

$$f(x) = \begin{cases} 3x^2 & , x \geq 1 \\ ax - 4 & , x < 1 \end{cases}$$

2.

$$f(x) = \begin{cases} \frac{4 \sin x}{x} & , x < 0 \\ a - 2x & , x \geq 0 \end{cases}$$

3.

$$f(x) = \begin{cases} 1 - 4x & , x < 2 \\ ax^2 - 3x + 2 & , x \geq 2 \end{cases}$$

1.3 Properties of Continuity

Theorem 1.3.1 If b is a real number and f and g are continuous at $x = c$, then the following functions are also continuous at c

1. Scalar multiple: bf
2. Sum or Difference: $f \pm g$
3. Product: fg
4. Quotient: $\frac{f}{g}$, if $g(c) \neq 0$.

Note: The following types of functions are continuous at every point in their domains.

- (i) Polynomials

- (ii) Rational $r(x) = \frac{p(x)}{q(x)}$, $q(x) \neq 0$
- (iii) Radical: $f(x) = \sqrt[n]{x}$
- (iv) Trigonometric: $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$

Exercise 1.3.1 Find the following:

1. Prove that $f(x) = \sin(ax)$, $(a \neq 0)$ is continuous on $\mathbb{R}$.

2. Prove that $f(x) = x^3 - x$ is continuous on $\mathbb{R}$.

3. Discuss the continuity of the function $f(x) = \ln(x^2 + 1)$

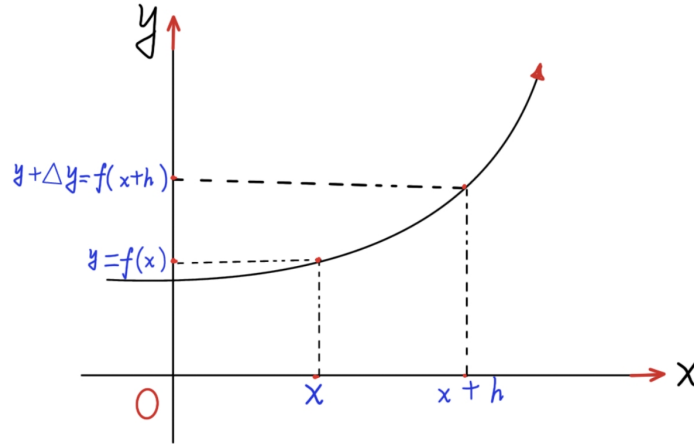
4. Determine whether the following functions are continuous at the given points.

$$(a) \ f(x) = \begin{cases} x^x - 1 & , x \neq 0 \\ 1 & , x = 0 \end{cases} \text{ at } x = 0$$

$$(b) \ f(x) = \begin{cases} \frac{\sin(\pi x)}{1-x} & , x < 1 \\ \pi x^2 & , x \geq 1 \end{cases} \text{ at } x = 1$$

2 The Differentiation

2.1 The Change and Rate of Change



Let $y = f(x)$ is a continuous function on an open interval I , $(x \in I)$.

1. The change on $x \equiv \Delta x = x_2 - x_1 = h$
2. The change on $y \equiv \Delta y = f(x+h) - f(x)$
3. The rate of change is $\frac{\Delta y}{\Delta x}$

$$\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}$$

Example 2.1.8 If $y = x^3$, find $\frac{\Delta y}{h}$

Sol

$$\begin{aligned} \frac{\Delta y}{h} &= \frac{(x+h)^3 - x^3}{h} = \frac{h[(x+h)^2 + x(x+h) + x^2]}{h} \\ &= (x+h)^2 + x(x+h) + x^2 = 3x^2 + 3hx + h^2 \end{aligned}$$

Therefore $\frac{\Delta y}{h} = \boxed{3x^2 + 3hx + h^2}$

2.2 Rate of Changes or Derivatives

Definition 2.2.3 The derivative of f is the function:

$$f'(x) = \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\Delta y}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (2.2.1)$$

provided the limit exists and x is in the domain of f . If $f'(x)$ exists, we say that f is **differentiable** at x .

Note: If f is differentiable at every point of an open interval I , we say that f is differentiable on I .

Example 2.2.9 Given $y = x^3$, evaluate the rate of change $\frac{dy}{dx}$.

Sol

Since $y = x^3$ is a polynomial $\Rightarrow$ continuous on $\mathbb{R}$. By using 2.2.1

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

from example 2.1.8, we saw that $\frac{f(x+h) - f(x)}{h} = 3x^2 + 3hx + h^2$

$$L.S.L = \lim_{h \rightarrow 0^+} (3x^2 + 3hx + h^2) = \lim_{h \rightarrow 0} (3x^2 + 3hx + h^2) = 3x^2.$$

$$R.S.L = \lim_{h \rightarrow 0^-} (3x^2 + 3hx + h^2) = \lim_{h \rightarrow 0} (3x^2 + 3hx + h^2) = 3x^2$$

Therefore $\boxed{\frac{dy}{dx} = 3x^2}$

Example 2.2.10 For the function $y = |x|$, find $\frac{dy}{dx}$ at $x = 0$.

Sol

We know that, the given function is continuous on $\mathbb{R}$.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - 0}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

Now, we will study the existence of the limit by calculating the left and right limits.

$$(i) L.S.L = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0} \frac{h}{-h} = -1$$

$$(ii) L.S.L = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

Thus the limit does not exist at $x = 0$ and here we say the function is not differentiable at $x = 0$.

Theorem 2.2.2 Differentiable implies continuous

Proof

Because f is differentiable at a point c , we know that:

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists. To show that f is continuous at c , we must show that $\lim_{x \rightarrow c} f(x) = f(c)$.

We can write $f(x)$ as:

$$f(x) = \frac{f(x) - f(c)}{x - c}(x - c) + f(c) \quad , x \neq c \quad (2.2.2)$$

By taking the limit as x approaches c on both sides of equation 2.2.2 and simplifying, we get:

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c}(x - c) + f(c) \right] \\ &= \underbrace{\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}}_{f'(c)} \times \underbrace{\lim_{x \rightarrow c} (x - c)}_0 + \underbrace{\lim_{x \rightarrow c} f(c)}_{f(c)} = f'(c) \times 0 + f(c) = f(c) \end{aligned}$$

Therefore $\lim_{x \rightarrow c} f(x) = f(c)$ which means that f is continuous at x .

Theorem 2.2.3 Not continuous implies not differentiable.

If f is not continuous at $x = c$, then f is not differentiable at $x = c$

2.3 Rules of Differentiation

1. The Constant function $y = c$.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

$$\boxed{\therefore \frac{d}{dx}[c] = 0}$$

2. Power Rule

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

Proof

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} = \lim_{h \rightarrow 0} \frac{[x^n + nx^{n-1}h + \binom{n}{2}n^{n-2}h^2 + \dots + h^n] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{h[nx^{n-1} + \binom{n}{2}n^{n-1}h + \dots + h^{n-1}]}{h} \\ &= \lim_{h \rightarrow 0} [nx^{n-1} + \binom{n}{2}n^{n-1}h + \dots + h^{n-1}] = nx^{n-1}. \end{aligned}$$

3. The Sine function $y = \sin x$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = \lim_{h \rightarrow 0} \frac{2 \cos\left(x + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)}{h} \\ &= \left[\lim_{h \rightarrow 0} \cos\left(x + \frac{h}{2}\right) \right] \left[\lim_{\frac{h}{2} \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right] = \cos x \times 1 = \cos x. \end{aligned}$$

$$\boxed{\therefore \frac{d}{dx}[\sin x] = \cos x}$$

4. The Cosine function $y = \cos x$

$$\frac{d}{dx}[\cos x] = -\sin x$$

Proof, left for you.

5. The Tangent Function $y = \tan x$

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h} = \lim_{h \rightarrow 0} \frac{\frac{\tan x + \tan h}{1 - \tan x \tan h} - \tan x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\tan x + \tan h - \tan x(1 - \tan x \tan h)}{h(1 - \tan x \tan h)} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{\tan x} + \tan h - \cancel{\tan x} + \tan^2 x \tan h}{h(1 - \tan x \tan h)} \\ &= \underbrace{\lim_{h \rightarrow 0} \left(\frac{\tan h}{h}\right)}_{(1)} \times \underbrace{\lim_{h \rightarrow 0} \left(\frac{1}{1 - \tan x \tan h}\right)}_{(1)} \times \underbrace{\lim_{h \rightarrow 0} (1 + \tan^2 x)}_{\sec^2 x} = 1 \times 1 \times \sec^2 x = \sec^2 x \end{aligned}$$

6. The Exponential function $y = a^x$, $a > 0$ ($a \neq 1$).

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

Now, let $\frac{1}{u} = a^h - 1 \Rightarrow u = \frac{1}{a^h - 1}$, thus $u \rightarrow \infty$ when $h \rightarrow 0$ and $h = \log_a \left(1 + \frac{1}{u}\right)$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= a^x \times \lim_{u \rightarrow \infty} \frac{1}{u \left(\log_a \left(1 + \frac{1}{u}\right)\right)} = a^x \times \lim_{u \rightarrow \infty} \frac{1}{\log_a \left(1 + \frac{1}{u}\right)^u} \\ &= a^x \times \frac{1}{\log_a \left(\lim_{u \rightarrow \infty} \left(1 + \frac{1}{u}\right)^u\right)} = a^x \times \frac{1}{\log_a e} = a^x \times \ln(a) \end{aligned}$$

$$\boxed{\therefore \frac{d}{dx}[a^x] = a^x \times \ln a}$$

7. The Logarithmic function $y = \log_a x$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\log_a(x+h) - \log_a x}{h} = \lim_{h \rightarrow 0} \frac{\log_a \left(1 + \frac{h}{x}\right)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \log_a \left(1 + \frac{h}{x}\right)$$

Let $u = \frac{x}{h} \Rightarrow u \rightarrow \infty$ when $h \rightarrow 0$ and $h = \frac{x}{u}$.

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{1}{h} \log_a \left(1 + \frac{h}{x}\right) = \lim_{u \rightarrow \infty} \frac{u}{x} \log_a \left(1 + \frac{1}{u}\right) = \frac{1}{x} \lim_{u \rightarrow \infty} \log_a \left(1 + \frac{1}{u}\right)^u \\ &= \frac{1}{x} \log_a \left[\lim_{u \rightarrow \infty} \left(1 + \frac{1}{u}\right)^u \right] = \frac{1}{x} \log_a(e) = \frac{1}{x \ln a} \end{aligned}$$

And we conclude that $\frac{d}{dx}[\ln x] = \frac{1}{x}$

Example 2.3.11 Find the derivative of the function $y = ax + b$ from the first principles.

Sol

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{a(x+h) + b - (ax + b)}{h} = \lim_{h \rightarrow 0} \frac{ax + ah + b - ax - b}{h} \\ &= \lim_{h \rightarrow 0} \frac{ah}{h} = \lim_{h \rightarrow 0} a = a. \end{aligned}$$

Example 2.3.12 Find the derivative of the function $y = 4 - 3x$ from the first principles.

Example 2.3.13 Find the derivative of the function $y = x^2 + 2x - 5$ from the first principles.

Sol

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 2(x+h) - 5 - (x^2 + 2x - 5)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{(x+h)^2 - x^2}{h} + \frac{2h}{h} \right] = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} + \lim_{h \rightarrow 0} 2 \\ &= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h} + \lim_{h \rightarrow 0} 2 = \lim_{h \rightarrow 0} (2x+h) + \lim_{h \rightarrow 0} 2 = (2x+0) + 2 = 2x+2. \end{aligned}$$

Example 2.3.14 Find the derivative of the function $y = \frac{1}{\sqrt{x}}$ from the first principles.

Sol

$$\begin{aligned} \frac{\Delta y}{h} &= \frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h} = \frac{\sqrt{x} - \sqrt{x+h}}{h(\sqrt{x}\sqrt{x+h})} \times \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} \\ &= \frac{x - (x+h)}{h(\sqrt{x}\sqrt{x+h})(\sqrt{x} + \sqrt{x+h})} = \frac{-1}{(\sqrt{x}\sqrt{x+h})(\sqrt{x} + \sqrt{x+h})} \\ \Rightarrow \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{-1}{(\sqrt{x}\sqrt{x+h})(\sqrt{x} + \sqrt{x+h})} = \frac{-1}{(\sqrt{x}\sqrt{x})(\sqrt{x} + \sqrt{x})} = \frac{-1}{x(2\sqrt{x})} = \frac{-1}{2x^{\frac{3}{2}}} \end{aligned}$$

2.4 Properties of Derivatives

1. **Constant multiple rule:** If f is differentiable at x and c is a constant, then

$$\frac{d}{dx}[cf(x)] = cf'(x)$$

2. **Sum Rule:** If f and g are differentiable at x , then

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$

Proof

Let $F = f + g$, where f and g are differentiable at x , so:

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) + g(x+h) - f(x) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= f'(x) + g'(x). \end{aligned}$$

Note:

- (a) The sum rule can be extended to three or more differentiable functions $f_1, f_2, \dots, f_n$ to obtain the generalized sum rule.

$$\frac{d}{dx}[f_1(x) + f_2(x) + \dots + f_n(x)] = f_1'(x) + f_2'(x) + \dots + f_n'(x)$$

- (b) The difference of two functions $f - g$ can be rewritten as the sum $f + (-g)$. By combining the sum rule with the constant multiple, the difference rule established.

$$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$$

The proof (exercise)

3. **Product Rule:** If f and g are differentiable at x , then

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Proof

Let $F(x) = f(x).g(x)$, then

$$\begin{aligned} \frac{dF}{dx} &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h)}{h} + \lim_{h \rightarrow 0} \frac{f(x)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\underbrace{\frac{f(x+h) - f(x)}{h}}_{f'(x), \text{ as } h \rightarrow 0} \cdot \underbrace{g(x+h)}_{g(x), \text{ as } h \rightarrow 0} \right] + \lim_{h \rightarrow 0} \left[\underbrace{f(x)}_{f(x), \text{ as } h \rightarrow 0} \cdot \underbrace{\frac{g(x+h) - g(x)}{h}}_{g'(x), \text{ as } h \rightarrow 0} \right] \\ &= f'(x)g(x) + f(x)g'(x) \end{aligned}$$

4. **Quotient Rule** Consider the quotient $q(x) = \frac{f(x)}{g(x)}$ and note that $f(x) = g(x).q(x)$. By the product rule, we have:

$$f'(x) = g'(x)q(x) + g(x)q'(x)$$

Solving for $q'(x)$, we find that

$$q'(x) = \frac{f'(x) - g'(x)q(x)}{g(x)}$$

Then substituting $q(x)$ with $\frac{f(x)}{g(x)}$ we get:

$$q'(x) = \frac{f'(x) - g'(x) \cdot \frac{f(x)}{g(x)}}{g(x)} = \frac{g(x)f'(x) - g'(x)f(x)}{(g(x))^2}$$

Theorem 2.4.4 Quotient Rule

If f and g are differentiable at x , $g(x) \neq 0$, then the derivative of $\frac{f}{g}$ at x exists and

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - g'(x)f(x)}{(g(x))^2}$$

Proof

$$\begin{aligned} \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - f(x)g(x+h)}{h(g(x)g(x+h))} \\ &= \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - f(x)g(x+h) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{h(g(x)g(x+h))} \\ &= \frac{\lim_{h \rightarrow 0} \frac{g(x)[f(x+h) - f(x)]}{h} - \lim_{h \rightarrow 0} \frac{f(x)[g(x+h) - g(x)]}{h}}{\lim_{h \rightarrow 0} (g(x)g(x+h))} \\ &= \frac{g(x) \lim_{h \rightarrow 0} \frac{[f(x+h) - f(x)]}{h} - f(x) \lim_{h \rightarrow 0} \frac{[g(x+h) - g(x)]}{h}}{\lim_{h \rightarrow 0} (g(x)g(x+h))} \\ &= \frac{g(x)f'(x) - g'(x)f(x)}{(g(x))^2} \end{aligned}$$

Note: $\lim_{h \rightarrow 0} g(x+h) = g(x)$ because g is given differentiable and therefore is continuous.

5. The Chain Rule

Theorem 2.4.5 If $y = f(u)$ is a differentiable function of u , and $u = g(x)$ is a differentiable function of x , then $y = f(g(x))$ is a differentiable function of x and

$$\boxed{\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)}$$

Example 2.4.15 Evaluate $\frac{dy}{dx}$ for the following functions.

1. $y = 4x^3 + \frac{3}{x} + 2\sqrt{x} - 7$

Sol

2. $y = 3 \cos x - \frac{1}{2} \sin x$

Sol

3. $y = x^7 + \frac{1}{3}x^{-2} - \frac{1}{\sqrt{x}} + \sqrt{x^3}$

Sol

4. $y = x \sin x$

Sol

5. $y = x^3 \cos x + x^2 - 5$

Sol

6. $y = (x + 4)(x^2 - x + 5)$

Sol

7. $y = \frac{3x^2 + 1}{6x + 5}$

Sol

8. $y = \frac{1 + \cos x}{1 - \sin x}$

Sol

9. $y = 2^{-x} + 3^{x+1}$

Sol

10. $y = \log_4(x) - 2 \ln x$

Sol

11. $y = x \sin(3x + 4)$

Sol

12. $y = \frac{\sin x + \cos x}{\sin x - \cos x}$

Sol

13. $y = \log_3 \left(x + \sqrt{x^2 + 1} \right)$

Sol

Exercise 2.4.2 1. From the first principles, evaluate the first derivative for the following function.

(i) $y = x^3$

(ii) $y = \frac{x}{x^2 - 5}$

$$(iii) \ y = \sin 2x$$

$$(iv) \ y = xe^{-x}$$

2. Find $\frac{dy}{dx}$ for the following:

$$(a) \ y = \sin(x \cos 2x)$$

$$(b) \ y = \frac{x^2 + ax + b}{x^2 - ax + b}$$

$$(c) \ y = \sqrt{1 + \sqrt{2^x - 1}}$$

$$(d) \ y = \sin^2 \left(\frac{2x}{x+1} \right)$$

$$(e) \ y = \frac{1 + 2 \tan^2 x}{1 - \tan^2 x}$$

$$(f) \ y = \ln(\sec x + \tan x)$$

$$(g) \ y = \frac{1}{\sqrt{1 + \sqrt{x}}}$$

$$(h) \ y = \frac{x^2}{\sqrt{a^2 - x^2}}$$

$$(i) \ y = xe^{-x^2}$$

$$(j) \ y = x \cos(\sin 2x)$$

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